

22nd Experiment

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import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import make_classification
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Generate Two-Class Dataset
X, y = make_classification(
    n_samples=500,
    n_features=2,
    n_redundant=0,
    n_informative=2,
    n_clusters_per_class=1,
    class_sep=2,
    random_state=42
)

# Build Neural Network
model = Sequential([
    Dense(8, activation='relu', input_shape=(2,)),
    Dense(4, activation='relu'),
    Dense(1, activation='sigmoid')
])

# Compile Model
model.compile(
    optimizer='adam',
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        loss='binary_crossentropy',
        metrics=['accuracy']
    )

# Train Model
history = model.fit(
    X, y,
    epochs=50,
    verbose=0
)

# Training Loss Graph
plt.figure(figsize=(5,3))
plt.plot(history.history['loss'])
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()

# Prediction
pred = model.predict(X)
pred_class = (pred > 0.5).astype(int)

# Accuracy
acc = accuracy_score(y, pred_class)
print("Accuracy:", round(acc*100,2), "%")

# Dataset Visualization
plt.figure(figsize=(5,4))
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plt.scatter(X[:,0], X[:,1], c=y)
plt.title("Two-Class Dataset")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```

